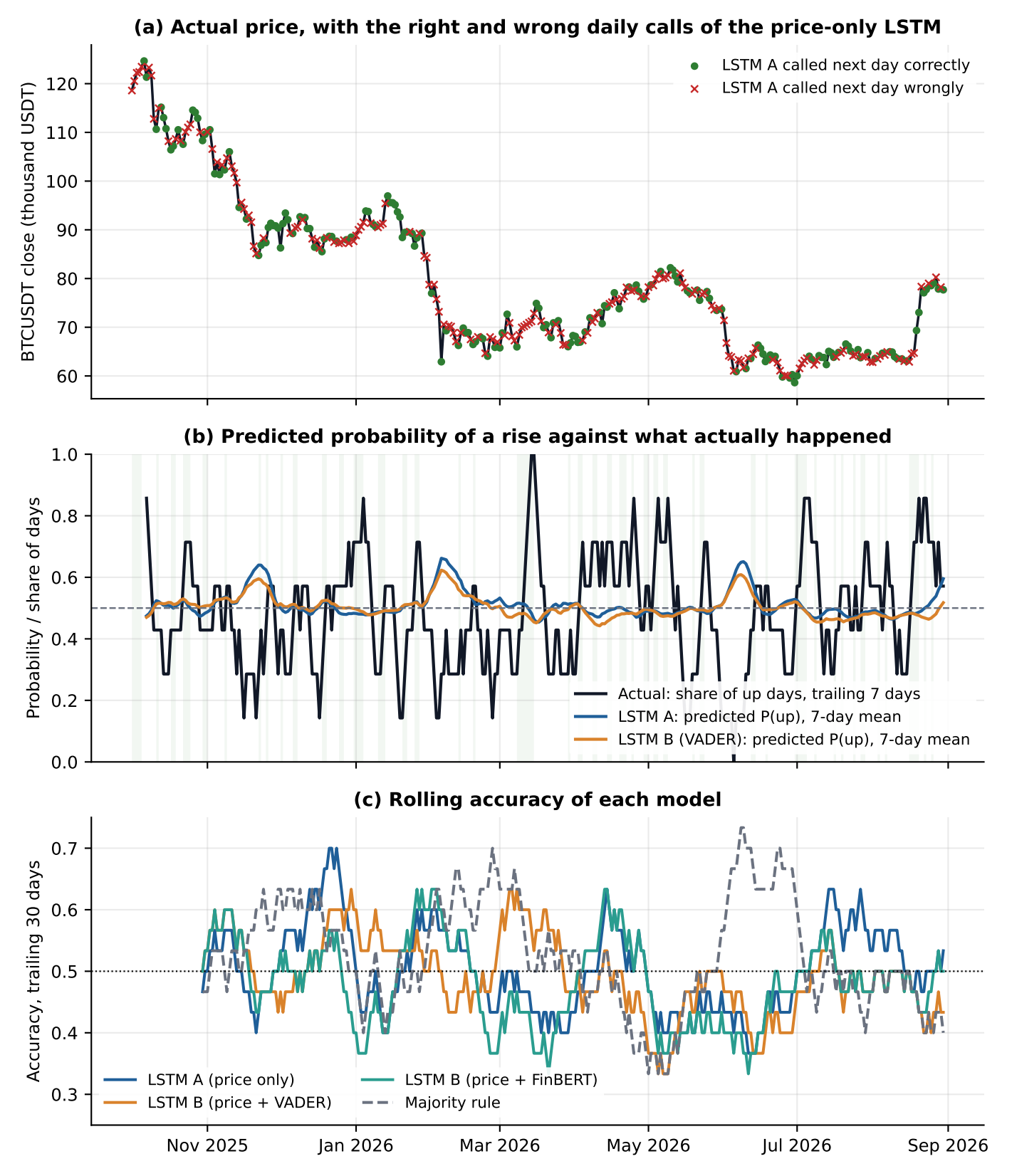
(a) Actual price, with the right and wrong daily calls of the price-only LSTM
BTCUSDT close (thousand USDT)
120
110
100
90
80
70
60
LSTM A called next day correctly
LSTM A called next day wrongly
(b) Predicted probability of a rise against what actually happened
Probability / share of days
1.0
0.8
0.6
0.4
0.2
0.0
Actual: share of up days, trailing 7 days
LSTM A: predicted P(up), 7-day mean
LSTM B (VADER): predicted P(up), 7-day mean
(c) Rolling accuracy of each model
Accuracy, trailing 30 days
0.7
0.6
0.5
0.4
0.3
LSTM A (price only)
LSTM B (price + VADER)
LSTM B (price + FinBERT)
Majority rule
Nov 2025
Jan 2026
Mar 2026
May 2026
Jul 2026
Sep 2026